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College of Computer and Information Sciences
Information Systems Department
Project

Course Title:	Requirements Engineering
Course Code:	IS 1130
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Instructor:	Male: Dr. Yasser Kotb
Semester:	2 nd semester 2024-25
Due Date:	26-1-2025

Instructions:

1. The assignment is to be carried out in groups of 3-4 students.
2. The submission date is 26 January **2025** via *Blackboard* before **11:59pm** Hard Copy Sunday 26/1/2025 Lecture time.
3. Any attempt to cheat from any of your classmates or form the Internet the whole assignment will be marked as **Zero**. Be aware that Blackboard uses plagiarism check tool! (Read and sign **Plagiarism Declaration**)

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Criteria	Student Marks	Question Marks
Report		15

Presentation		5
Total		20

Project Evaluation's Sheet

Report	Total 15	
(CLO 3.1)	8	
Criteria	Deserved Points	Criteria's Points
• Project Scope and Business Objective		1
• Identifying Stakeholders		1
• Using elicitation techniques		1
• Functional and non-functional Requirements (at least 30)		5
(CLO 2.2)	7	
• Use Case Diagram and User requirements		2
• Activity Diagram		2
• Requirement Validation		2
• Report Format		1
SubTotal#1		15

Presentation (CLO 3.2)	5	
Criteria	Deserved Points	Criteria's Points
• Slides		2.5
• Discussion		2.5
SubTotal#2		5

Total	20	
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1. Introduction

1.1 Purpose

The purpose of the "AI chatbot" is to ease the process of seeking help through customer support.

This tool is beneficial for customers by being efficiently available 24/7.

In addition, it cut costs effectively whilst ensuring customer's satisfaction

1.2 References

- **ISO/IEC/IEEE 29148-2011: Systems and software engineering — Life cycle processes — Requirements engineering.**
- **IS1130 Course Materials: Requirements engineering course materials from Imam Mohammad Ibn Saud Islamic University.**
- **Salesforce Title: *How AI Chatbots Improve Customer Experience and Operational Efficiency* Publisher: Salesforce Description: A case study that explores how Salesforce AI chatbots have transformed customer support, focusing on efficiency and user satisfaction. Link: <https://www.salesforce.com>**

2. Description

2.1 Product Perspective

The AI chatbot system is a tool that is used to initiate conversations between the organization's customers support and the users to help guarantee their satisfaction in any given moment of the day.

3. Project Scope and Business Objective:

3.1 Scope:

3.1.1. An AI chatbot for customer support is a system that talks to customers online. Its scope to help answer questions quickly and easily, making it simpler for customers to get the help they need. The AI chatbot can handle common issues, give information about products, and help solve basic problems, this way, customer doesn't have to wait long and support staff can focus on complex tasks.

3.1.2. Out Of Scope:

3.1.3. AI chatbot cannot simulate human emotions

3.1.4. AI chatbot cannot develop itself. Developers must intervene

3.1.5. AI chatbot cannot repair system malfunctions

3.2 Business objectives of AI Chatbot for customer support:

3.2.1. To improve customers journey by providing instant and accurate responses to queries, making sure customers know that they are being heard and valued

3.2.2. This automation will be a way of improving efficiency, that is, by reducing operational costs, allowing human support staff to focus on more complex issues and improving overall efficiency

3.3.3. To increase availability and allow customers to receive assistance at any time, day or night, regardless of regular business operating hours

4. Identifying Stakeholders:

4.1 Customers or Clients:

Customers have a direct interest in this project, and they're the main stakeholders of the AI chatbot system. They're seeking assistance from the system, finding information. And the chatbot is designed to assist them as fast as possible at any time during the day to ensure the satisfaction of the client.

4.2 System developers:

System developers have a direct interest in this project, they're responsible for designing /developing and maintaining the system. The chatbot operates smoothly, connects users to the CRM, and gets updated regularly, training the AI is all their main priority.

4.3 The business owners/managers:

The business owner has an indirect interest in this project, the manager of the organization that uses the system is concerned with the user satisfaction with how fast and responsive the chatbot is, and how it will cut costs. So, they are interested in how efficient the system is and the cost effectiveness. The AI can help achieve the organization's goals.

5. Elicitation Techniques:

5.1 Interviews

Description:

Interviews describe personal discussions directly with the stakeholders to get minute details about their expectations and requirements.

One-on-one or small-group interviewing.

Application to the Project:

Stakeholders: customer support managers, developers of the chatbot, and end-users, who are the customers.

Questions to Ask:

What is the most common issue customer support teams face?

What kind of queries do customers ask repeatedly?

What functionalities/features should be included in the chatbot to enhance the user experience?

Outcome:

Insights into main features: how queries are dealt with, language support, and processes for escalation.

5.2 Questionnaires

Description:

Questionnaires are distributed to get the response from more people. It is a structured way of gathering data.

Make sure the response options cover all the logical possibilities.

Application to the Project:

Target Audience: Customers availing themselves of customer support services of STC, Spotify, noon, and SPL Amazon.

Questions to Include:

Rate the present chatbot using a scale of 1–5 or emoji ratings.

Which of the following features in a chatbot helps you most?

Which changes do you want to be included in a customer care chatbot?

Output:

A dataset identifying the level of customer satisfaction and desirable features.

5.2.1. Questionnaire for AI Chatbot Customer Support

We conducted a questionnaire for the consumer with these questions:

- 1. How would you rate your experience with the current AI chatbot?**
 - a. 1 (Poor)
 - b. 2 (Good)
 - c. 3 (Excellent)
- 2. Which feature do you find most useful in the chatbot?**
 - a. Instant responses to FAQs
 - b. Order tracking
 - c. Escalation to a live agent
- 3. What additional features would you like in the chatbot?**
 - a. Personalized recommendations
 - b. Voice interaction support
 - c. Multilingual capabilities
- 4. How do you prefer to interact with the chatbot?**
 - a. Text-based chat
 - b. Voice-based interaction
 - c. Both
- 5. Do you want the chatbot to provide updates on your queries automatically?**
 - a. Yes
 - b. No
- 6. Would you like to have a feedback option after each interaction?**
 - a. Yes
 - b. No
- 7. Do you need the chatbot to save chat history for future reference?**
 - a. Yes
 - b. No
- 8. What improvements do you want in the chatbot?**
 - a. Faster response time
 - b. More accurate responses
 - c. Easier navigation
- 9. Would you use a chatbot that supports your native language?**
 - a. Yes
 - b. No**

<https://docs.google.com/forms/d/e/1FAIpQLSefu4b6wDT4rCYaAz0g1QnfonBgV6zHNeLIR1wT378qvBCZTQ/viewform?usp=header>

5.3 Observation

Description & Application in the Project:

Focus Areas:

Observe customer service representatives processing queries not managed by the chatbot.

Analyze customer interaction with the chatbot, for example, unresolved queries or escalation rates.

Validate data gathered from other sources.

Outcome:

Identify repetitive tasks suitable for automation.

Understand when and why customers need live agent support.

6. Functional and non-functional Requirements:

6.1 Customers or Clients (Direct)

Functional Requirements:

6.1.1. - The chatbot shall respond to customer queries in real time.

6.1.2. - The chatbot shall handle multiple languages to help many types of users.

6.1.3. - The chatbot shall do transactions such as order placement or cancellations or provide tracking info.

6.1.4. - It shall allow customers to upload files or documents for specific queries.

6.1.5. - Customers shall be able to rate the chatbot performance after an interaction.

Non-Functional Requirements:

6.1.6. - The response time of the chatbot shall not exceed 3 seconds.

6.1.7. - It shall be available with a minimum uptime of 99%.

**6.1.8. - The data exchanged between customers and the chatbot shall be encrypted.
for security.**

6.1.9. - The interface shall be smooth and user-friendly for all age groups.

6.1.10. - The chatbot shall not conflict with the culture and tradition of the user.

6.2 System Developers and Engineers (Direct)

Functional Requirements:

- 6.2.1. - The developers shall have a dashboard to monitor the performance of chatbots.**
- 6.2.2. - The system shall allow easy updates to the training data for developers.**
- 6.2.3. - It shall integrate easily with the existing CRM and ticketing systems.**
- 6.2.4. - The system shall support API integrations for third-party tools.**
- 6.2.5. - The system shall provide analytics on user interactions to help the developers in its optimization.**

Non-Functional Requirements:

- 6.2.6. - The system shall be scalable to handle growing customer usage.**
- 6.2.7. - Codebase shall follow best practices for maintainability and readability.**
- 6.2.8. - The chatbot shall use the least possible computational resources.**
- 6.2.9. - Updates shall be rolled out seamlessly without disrupting the service.**
- 6.2.10. - The system shall support modular design to allow easy addition of features.**

6.3 Business Owners/Managers (Indirect)

Functional Requirements:

- 6.3.1. - Business managers shall have a dashboard for visualizing key metrics such as user satisfaction and cost savings.**
- 6.3.2. - The chatbot shall capture and store data on common customer pain points.**
- 6.3.3. - It shall send notifications to managers about critical issues.**
- 6.3.4. - It shall provide ROI metrics that justify cost savings and efficiency.**
- 6.3.5. - It shall send notifications for unresolved customer issues that are out of a predefined scope.**

Non-functional Requirements:

- 6.3.6. - The manager dashboard shall load within 2 seconds.**
- 6.3.7. - Reports generated by the system shall be visually clear and concise.**
- 6.3.8. - The chatbot shall increase support time response and satisfaction rate by 90%.**
- 6.3.9. - All interaction data shall be stored securely for at least 5 years.**
- 6.3.10. - Operation of the chatbot shall reduce support costs by at least 30% in the first year.**

Functional Requirements	Non-Functional Requirements
- It shall real-time responses to queries. - It shall handle multiple languages. - It shall support transactions (e.g., order placement, cancellations, tracking). - It shall allow file/document uploads for specific queries. - It shall enable rating of chatbot performance. - It shall dashboard to monitor chatbot performance. - It shall support easy updates to training data. - It shall integrate with existing CRM and ticketing systems. - It shall provide analytics on user interactions. - It shall support API integrations for third-party tools. - It shall dashboard for key metrics (e.g., satisfaction, cost savings). - It shall capture and store data on customer pain points. - It shall notify managers about critical issues and unresolved queries. - It shall provide ROI metrics for cost savings.	- It shall response time ≤ 3 seconds. - It shall be uptime $\geq 99\%$. - It shall provide data encryption for security. - It shall provide smooth, user-friendly interface for all age groups. - It shall align with user culture and tradition. - It shall scalability for growing usage. - It shall provide maintainable and readable codebase. - It shall be efficient use of computational resources. - It shall be seamless updates without service disruption. - It shall modular design for adding features. - It shall have dashboard loads within 2 seconds. - Reports shall be clear and concise. - It shall increase response time and satisfaction rate by 90%. - It shall store interaction data securely for 5+ years. - It shall reduce support costs by $\geq 30\%$ in the first year.

6. Table 1: Functional Requirements & Non-Functional Requirements

7. User requirements:

7.1 Customer/Client Requirements:

- 7.1.1. Shall provide chat support to the customer on a 24/7 basis.**
- 7.1.2. it Shall Responses should be faster to frequently asked questions such as product details.**
- 7.1.3. It shall provide multilingual support to various groups of users.**
- 7.1.4. It shall transfer to a human operator if necessary, seamlessly.**
- 7.1.5. It shall Secured and private with respect to customer data. Accessed easily through the web, an app, etc.**
- 7.1.6. Chatbot shall give answers exactly to queries in an appropriate manner.**
- 7.1.7. The system shall have a user-friendly environment of development to create and train chatbot solutions.**
- 7.1.8. Then the chatbot shall be interfaced with existing CRM tools and databases.**
- 7.1.9. The developers shall have an easy access to the log files and analytics modules for performance analysis of the system.**
- 7.1.10. Additionally, it shall do continuous improvement, including updates and training, must be supported by the system needed for the AI model.**
- 7.1.11. is shall will require documentation for all system components and APIs in detail.**
- 7.1.12. It shall also have features for easy debugging and troubleshooting in case of issues in the chatbot.**
- 7.1.13. The chatbot system also shall permits various customizations of their responses and workflows.**
- 7.1.14. it shall Provide testing tools for simulation to represent how a user interacts with the system while maximizing quality.**

7.2 Requirements of Users:

7.2.1. The chatbot shall produce all the analytical details regarding customer interactions and satisfaction levels.

7.2.2. This system shall reduce the cost associated with customer support systems.

7.2.3. It shall have support scalability to accommodate increases in the number of customer queries received.

7.2.4. It shall minimize a response time to achieve better satisfaction amongst customers.

7.2.5. Business administrators shall have a dashboard to get an overview of the efficiency of the bot.

7.2.6. This system shall have been made to comply with all relevant industrial as well as master standards.

8. Modelling diagram:

Use Case Diagram & Activity Diagram:

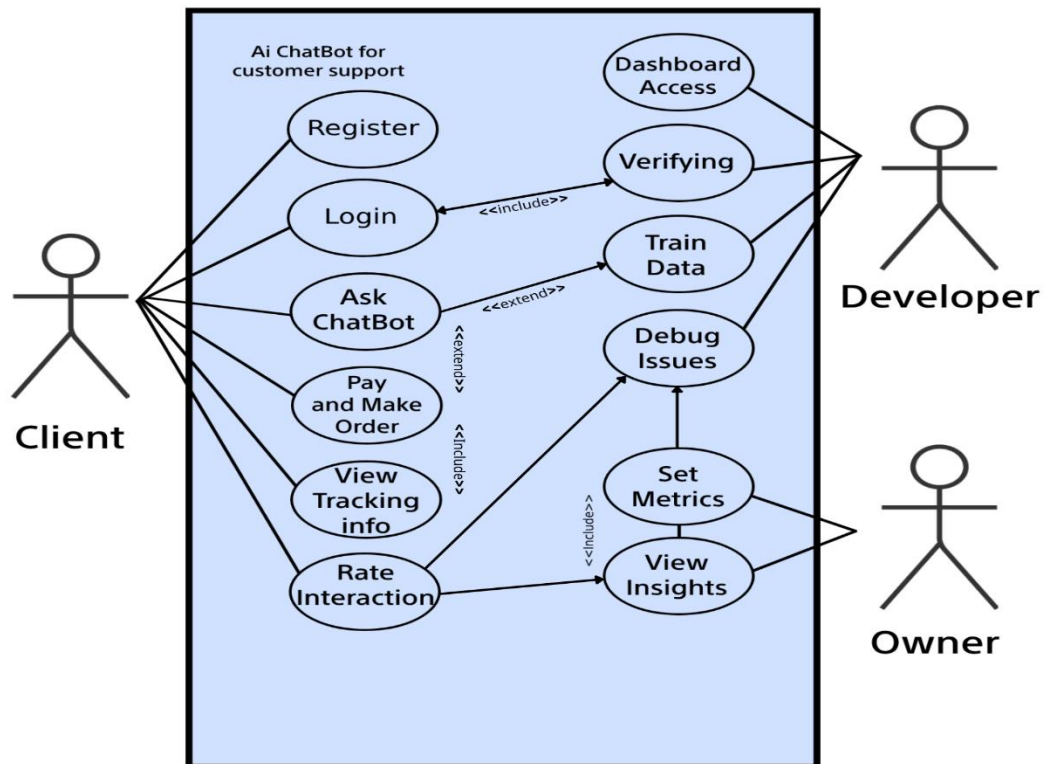
8.1 Use Case Diagram:

The Case Diagram depicts the interaction of various stakeholders (called actors) with the system. It also shows the various functionalities provided to each stakeholder group.

Actors and List of Use Cases :

Customers/Clients: Operate the Chatbot, Rate the Interaction

System Developers and Engineers: Update Training Data for the Chatbot, Debug Issues.



Figur1 The use case for relevant requirements

8.2 Activity Diagram:

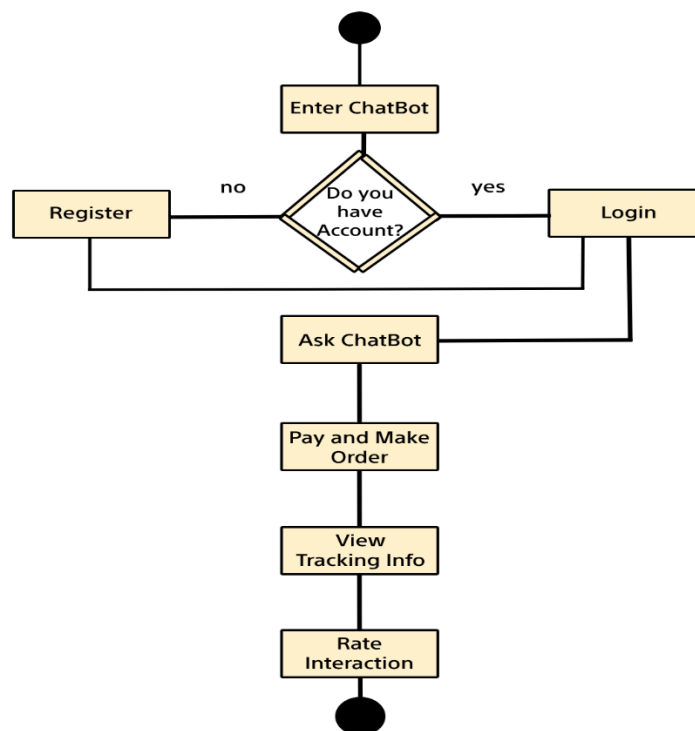
The selection of an Activity Diagram can enable the AI chatbot system to comprehend various user interactions that a system goes through besides other various system inputs and processes in a more concise way. An Activity Diagram can be better for the following reasons:

8.2.1. Process Notation

8.2.2. Dynamic Behaviour

8.2.3. Clear User Flow

8.2.4. Ease in Understanding



Figur2 The activity diagram

9. Requirement Validation:

9.1 Completeness Check

9.1.1. Functional Requirements: Every single functional requirement was well-defined, including those, which were paramount to the chatbot success for customers, developers, and business managers.

9.1.2. This includes real-time responses, multilingual support, integration with CRM tools, and performance monitoring.

9.1.3. The requirements for handling transactions, file uploads, and escalation to human agents are in place. That way, complete support for customer needs is guaranteed. Developer tools for dashboard monitoring, training data updates, and API integrations are specified.

9.1.4. Thus, technical maintenance is covered. Business manager's needs for analytics, notifications, and ROI metrics are met.

9.1.5. All functional requirements that had been identified were tested and all the correspondence about the chatbot's main functions for customers, developers, business managers, and every function was approved. The automatic response system, multilingual support, the integration of CRM tools, and the performance monitoring make up the functional requirements for the chatbot.

9.1.6. Transaction handling, file upload, and escalation to human agents shall be included. This will ensure that customer needs are taken care of by pointing out the details. Developer tools for dashboard monitoring, training data updates, and API integrations are comprehensively laid out. This helps in technical maintenance. Business managers' needs for analytics, notifications, and ROI metrics are well-fulfilled.

9.1.7. Developer tools for dashboard monitoring, training data updates, and API integrations are specified, covering the technical maintenance aspect.

Business managers' needs for analytics, notifications, and ROI metrics are well-covered.

9.1.8. Non-Functional Requirements: Non-functional parts such as response time, availability, scalability, data security, and system modularity are clearly described to all stakeholders. Compatibility and user-friendliness concerns, which are very important for the common people and diverse audiences, are also being discussed, so the system can be accessed and used without difficulty.

9.1.9. Quantity characteristics like response time, availability, scalability, data security and system modularity is a set of numbers known well by each member of the team.

Such approach cultural and user-friendliness, it provides for usability of the whole target audience.

9.2 Feasibility Check

9.2.1. Technical Feasibility: The requirements of AI chatbot technologies, such as NLP (Natural Language Processing) and CRM and ticketing system capabilities, correspond to the status quo. In cloud-based infrastructures and modern development practices, these demands can be addressed through scalable and modular designs. Developers' requirement for easy debugging, logs, and testing tools can be satisfied by the use of already developed development frameworks and tools.

9.2.3. The requirements of AI chatbot technologies, such as NLP (Natural Language Processing) and integration capabilities with CRM and ticketing systems, are in line with the current state of the technology.

9.2.4. The requirements for scalability and modular design can be met with the use of cloud technologies as well as modern development practices.

9.2.5. The requirement for developers to debug easily, to store logs, and to run tests can be made by the use of the already developed development frameworks and tools.

Operational Feasibility: In order to make this possible, stable and reliable infrastructure and NLP frameworks such as Google Dialogflow or Microsoft Bot Framework must be deployed in real time (99% uptime), and they should be able to support multiple languages. People can save on the expenses that come with running a chatbot by implementing these measures and concurrently satisfy customers which represents a benchmark for business.

9.2.6. In order to achieve the real-time availability (99% uptime) and the multilingual support the chatbot can use the robust infrastructure and different NLP frameworks like Google Dialogflow or Microsoft Bot Framework.

9.2.7. When it comes to operational feasibility, the key is the application of real-time availability (99% uptime) and the multilingual support through the implementation of stable infrastructure and NLP frameworks like Google Dialogflow or Microsoft Bot Framework.

9.2.8. Economic Feasibility: Automation will reduce support costs by at least 30% and ensure that the ROI metrics are met, therefore, this project is justified.

Automation will reduce support costs by at least 30% and ensure that the ROI metrics are met, therefore, this project is justified.

9.3 Consistency Check

9.3.1. Alignment with Stakeholders: Customer requirements determine quick, safe and smooth assistance, optimizing business objectives of user satisfaction improvement as it is.

Developers also share the same desire for flexibility, debugging, and performance monitoring for the reliability and maintainability integers of the system required by the developers. Business owners demand are high-level analytical analysis, cost-effectiveness, and well efficiency of the operational system, which are strategic organizational objectives affordable by implementation. Customer requirements determine fast, secured, and seamless support enabling the business goals of the user satisfaction improvements to be met. Developers also share the same desire for flexibility, debugging, and performance monitoring for the reliability and maintainability.

9.3.2. option of the system. Business owners demand are high-level analytical analysis, cost-effectiveness of non-reproducible machine failure and energy efficiency of the operating system. Conflict Resolution: There are no conflicting requirements detected; the needs of all stakeholders are in line with each other.

9.3.3. For instance, working cost reduction for administrators does not go against the quality of help towards customers. There are no conflicting requirements noticed; all interest groups' needs are complementary. For example, there is no issue of sacrificing the quality of customer support by reducing operational costs for administrators.

9.4 Verifiability

9.4.1. All requirements can be measured and tested:

All these requirements have to be checked through performance testing tools: Response time ≤ 3 seconds, uptime $\geq 99\%$, scalability.

9.4.2. Customer satisfaction: The chatbot will have various survey, ratings, and feedback mechanisms to calculate it.

9.4.3. System logs and analytics can verify the implementation of monitoring and optimization features for developers.

9.5 Traceability

9.5.1. Every one requirement can refer back to the requirements of the specific stakeholder needs. For example "the chatbot must respond in real-time" talks about the need for customers to receive instant help. API integration is the developers' need for easy integration of tools. Business owners need to generate ROI metrics to show the business case for expenditure.

9.6 Stakeholder Validation

9.6.1. Validation through the methods of elicitation (interviews, questionnaires, and observations) will guarantee that requirements have integrity with the stakeholder expectations:

Interviews: Stakeholder insight was used for the main priorities such as real-time response, escalation, and analytics.

9.6.2. Questionnaires: Customer preference for multilingual support as well as interface usability.

9.6.3. Observations: Resolve queries that validate the requirement for a handoff to the human agent and escalation.

9.6.4. Requirement Validation Outcome

This essentially defines requirements for the AI chatbot system:

complete coverage on all functional and non-functional fronts,

feasibility-in-technical, operational, and economic eyes,

State all such with respect to stakeholder's needs and organizational goal.

9.6.5. Verification: and readability: easy implementation and testing.

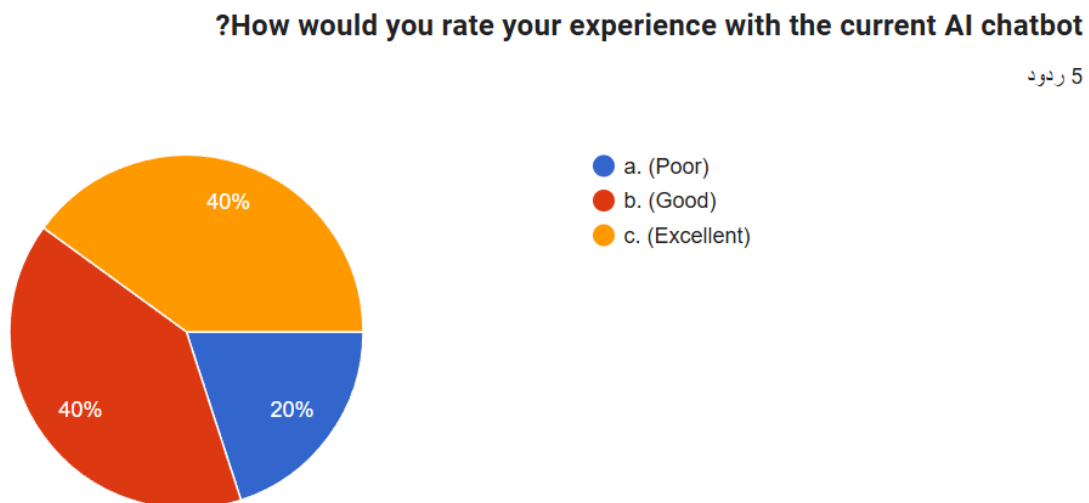
9.6.6. Maintaining traceability to the specific stakeholder need cannot be done without compromising clarity and focus.

9.6.7. This validation ensures a strong foundation for building the AI chatbot system and meeting initial stakeholder expectations.

Appendix A: Glossary:

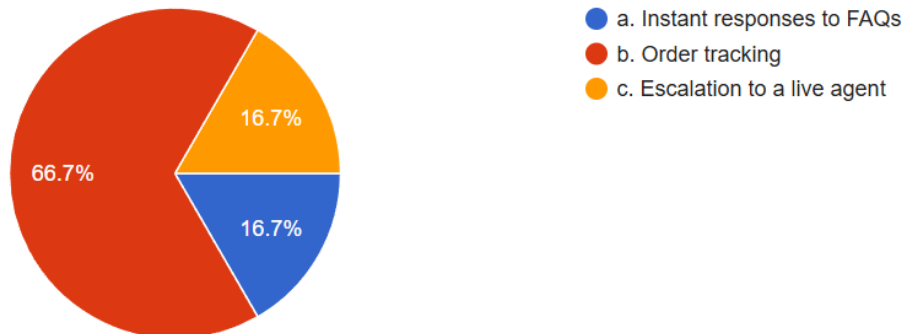
- **AI Chatbot:** An artificial intelligence-powered conversational agent designed to assist customers by answering queries, solving issues, and providing support.
- **CRM (Customer Relationship Management):** A system used to manage a company's interactions with current and potential customers, often integrated with the chatbot for seamless support.
- **Feedback System:** A feature allowing customers to rate their experience and provide comments after interacting with the chatbot.

Appendix B: Customer satisfaction Analysis model:



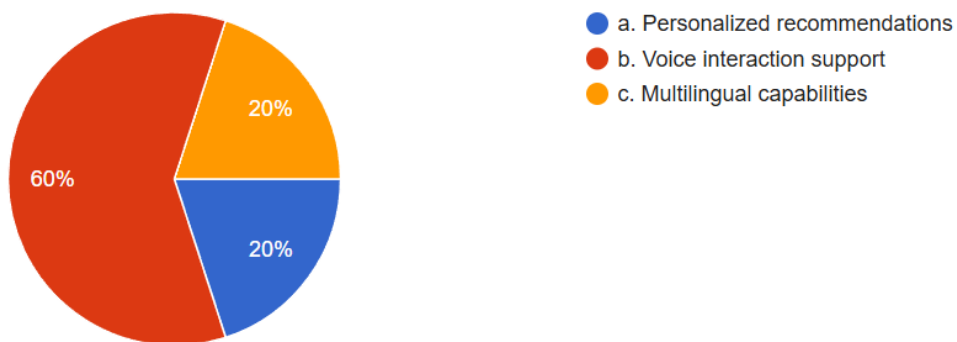
?Which feature do you find most useful in the chatbot

ردود 6



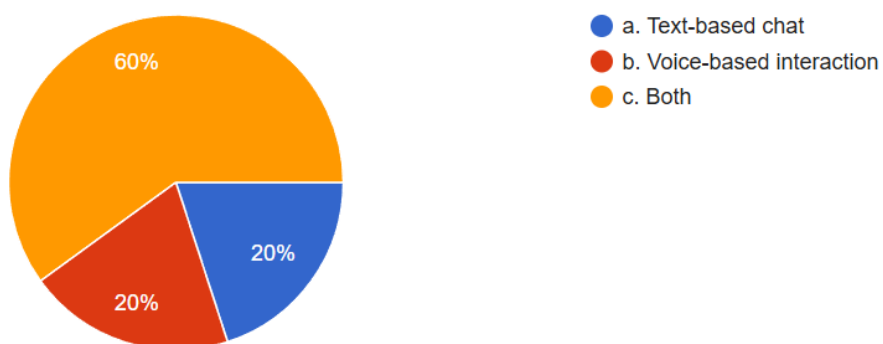
?What additional features would you like in the chatbot

ردود 5



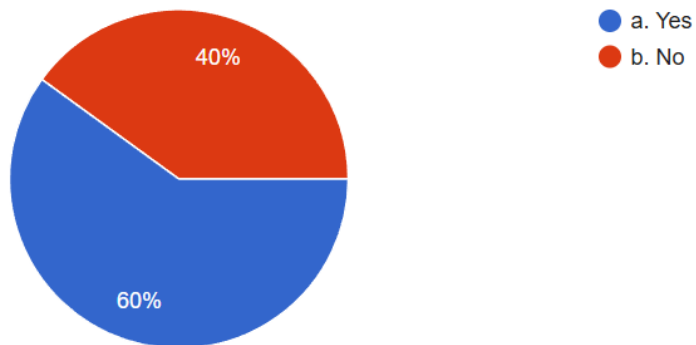
?How do you prefer to interact with the chatbot

ردود 5



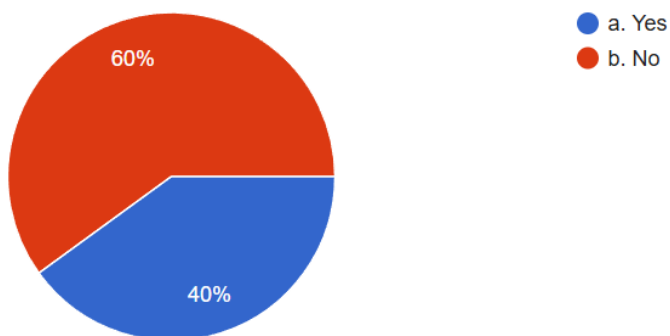
**Do you want the chatbot to provide updates on your queries
?automatically**

5 ردود



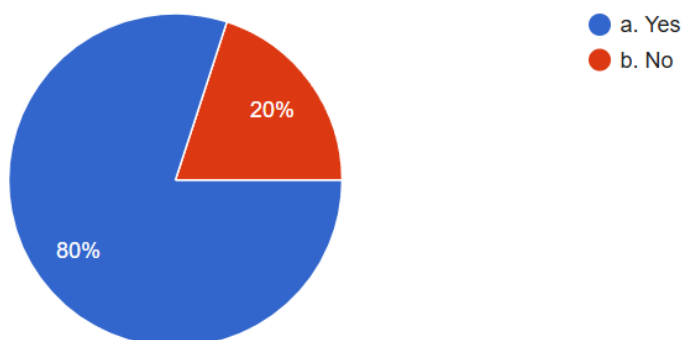
?Would you like to have a feedback option after each interaction

5 ردود



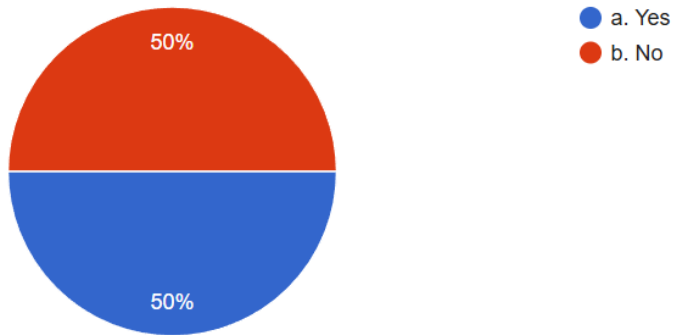
?Do you need the chatbot to save chat history for future reference

5 ردود



?Would you use a chatbot that supports your native language

6 ردود



?Which feature do you find most useful in the chatbot

6 ردود

